

Smart Exam Malpractice Detection System

Problem Statement

Online examinations are becoming increasingly common in educational institutions and recruitment processes. However, maintaining exam integrity is a major challenge due to cheating methods such as hidden electronic devices, impersonation, external assistance, suspicious browser activity, and unauthorized communication. Traditional online proctoring systems rely heavily on human invigilators, which can be time-consuming, expensive, and less effective in monitoring multiple candidates simultaneously.

There is a need for an intelligent AI-based system that can automatically monitor and analyze student behavior during online examinations using multiple input sources such as webcam video, microphone audio, typed responses, and screen activity to detect suspicious activities in real time.

Proposed Solution

The proposed system is a Multimodal AI-Based Smart Exam Malpractice Detection System designed to ensure fairness and transparency during online examinations. The system uses Artificial Intelligence, Computer Vision, Speech Analysis, and Machine Learning techniques to monitor candidates automatically.

The system analyzes: **Webcam Video:** Detects face movement, multiple persons, eye movement, and suspicious behavior. **Audio Monitoring:** Identifies background voices, conversations, or external assistance. **Typed Responses:** Detects unusual typing patterns and copied content. **Screen/Image Activity:** Monitors tab switching, unauthorized applications, and suspicious screen behavior. All collected data is combined using multimodal AI analysis to calculate a malpractice risk score. If suspicious activity is detected, the system generates real-time alerts for exam administrators and stores evidence for further review.

The system helps educational institutions conduct secure online examinations with reduced manual monitoring and improved accuracy in detecting malpractice activities.

Technologies Used

Technology	Purpose
Python	Backend Development
OpenCV	Video Monitoring
YOLO	Object Detection
Speech Recognition	Audio Analysis
NLP	Text Analysis
Flask / FastAPI	Web Application Backend
Machine Learning	Suspicious Behavior Detection